

Daksh Adhar

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EDUCATION

Carnegie Mellon University - School of Computer Science

Master of Science in Robotic Systems Development (MRSD) | GPA: 4.15/4

Pittsburgh, PA

May 2026

Indian Institute of Technology, Guwahati

Bachelor of Technology in Engineering Physics | GPA: 8.27/10.00

Guwahati, India

May 2024

Courses: Deep RL and Control, Learning for 3D Vision, Computer Vision, Optimal Control, Robot Mobility, Robot Autonomy

SKILLS

Python, C/C++, MATLAB, Julia, PyTorch, Tensorflow, MuJoCo, Isaac Gym, OpenAI Gym, ROS2, MoveIt2, Gazebo, PyBullet

EXPERIENCE

1X Technologies

Palo Alto, CA

AI Resident, Reinforcement Learning Team

May 2025 – August 2025

- Trained RL policies for dexterous manipulation on NEO hands, and added randomization for **sim-to-real** transfer
- Designed metrics to benchmark policies on **Isaac Gym** and **MuJoCo**, evaluating **sim-to-sim** robustness
- Developed a **ROS 2 Humble** C++ controller to deploy evaluated policies in real-time simulation and teleoperation
- Built Tkinter-based local and Streamlit-based browser app for object segmentation mask data collection using **SAM2**
- Integrated Cloudflare R2 and DBaver **SQL** backend to load frames and store operator clicks on 1M+ frames

Biomimetic Robotics & Artificial Intelligence Laboratory

Guwahati, India

Research Assistant, Prof. Shyamanta M. Hazarika

January 2023 – May 2024

- Built a testing framework in **PyBullet** and **OpenAI Gym** to train RL policies for bionic hand power grasping
- Formulated reward functions and used **Soft Actor-Critic** algorithm to enable grasp-and-lift of deformable objects
- Applied **domain randomization** for sim-to-sim, achieving 38% slip reduction and 14% decrease in deformation

Invention Factory, IIT Gandhinagar

Gujarat, India

Summer Internship, Prof. Nithin V. George

May 2022 – July 2022

- Built a Raspberry Pi wearable device to improve spatial hearing performance for hearing-impaired individuals
- Utilized a **TD0A** Algorithm with a 4-microphone setup to achieve directional sound detection with 10° resolution
- Integrated haptic motors to convey sound directions as tactile feedback, reducing user response time

PROJECTS

AR assisted Robotic Total Knee Arthroplasty | [Website](#)

October 2024 – December 2025

- Developed a surgical robot for total knee replacement, achieving **2 mm and 2°** accuracy in drilling surgical pins
- Utilized **DINOV3** for bone cluster detection, **SAM** for segmentation, and **ICP** for real-time surgical plan registration
- Designed a **ROS2/MoveIt2** planning subsystem for pose-based drilling and collision aware trajectory planning
- Implemented an **Apple Vision Pro** AR interface for real-time drill site visualization and complete system control

Combining Specialist Policies for Generalist Multi-Agent Play | [Code](#) | [Report](#)

September 2025 – December 2025

- Engineered a **Mixture-of-Experts** using a gating network to compose specialist policies from shared latent embeddings
- Implemented distributed **self-play** using **IMPALA** with PPO to stabilize learning against non-stationary opponents
- Outperformed specialist and generalist baselines achieving 2nd place by gating policies to improve gameplay

Unitree G1 Humanoid Soccer Ball Kicking | [Code](#) | [Report](#)

January 2025 – April 2025

- Simulated penalty-style ball kicking on the Unitree G1 humanoid with whole-body motion and balance control
- Computed strike impulse using quadratic programming (**QP**) and joint trajectories through **DIRCOL** optimization
- Stabilized the robot using **IHLQR** and tracked kicking motion with **TVLQR**, achieving <0.5 m error from targets

Pixel art using Franka Emika Panda Arm | [Code](#) | [Report](#)

January 2025 – April 2025

- Developed a Franka Emika Panda pixel-art stamping robot with **position** and **force control** for precise stamping
- Optimized stamping sequence as a travelling salesman problem (**TSP**) using **Christofides** algorithm

PUBLICATIONS

CMU Vision-Language-Autonomy Challenge (Presentation)

October 2025

IROS 2025; AI Meets Autonomy: Vision Language and Autonomous Systems Workshop

[IROS Workshop](#)

Grasp force optimization as a Bilinear Matrix inequality problem: A Deep-learning approach

December 2023

6th National Conference on Multidisciplinary Design, Analysis and Optimization

[arXiv/2312.05034](#)

Reinforcement Learning-Based Bionic Reflex Control for Anthropomorphic Robotic Grasping

November 2023

Journal Paper at Cambridge University Press, Robotica

[Cambridge/Robotica](#)